TOPIC 6

DOMAIN APPLICATION MODELING

UTM-SERG Research & Consultancy Team

School of Computing

Faculty of Engineering

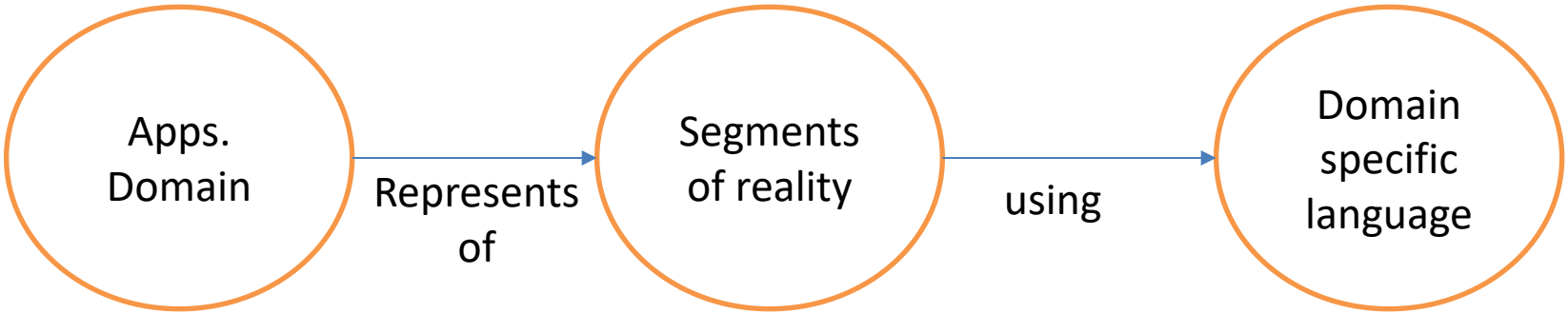
Universiti Teknologi Malaysia
UTM, Skudai, Johor.

AGENDA

- Agenda 1: Application Software and Domain
- Agenda 2: Identification of Domain Concepts
- Agenda 3: A Language based approach to concept design
- Agenda 3: Domain Driven Design and CRC Card
- Agenda 4: Conclusion

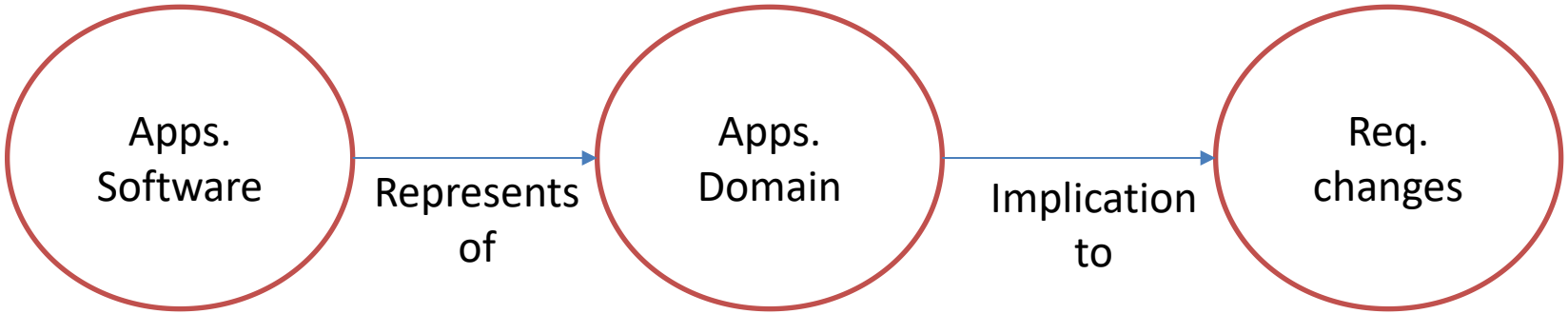
APPLICATION SOFTWARE AND DOMAIN

Application Domain – A segment of reality (real-world)



APPLICATION SOFTWARE AND DOMAIN

Application Software – To supports domain specific tasks and changing In the domain will forcing a change in a software



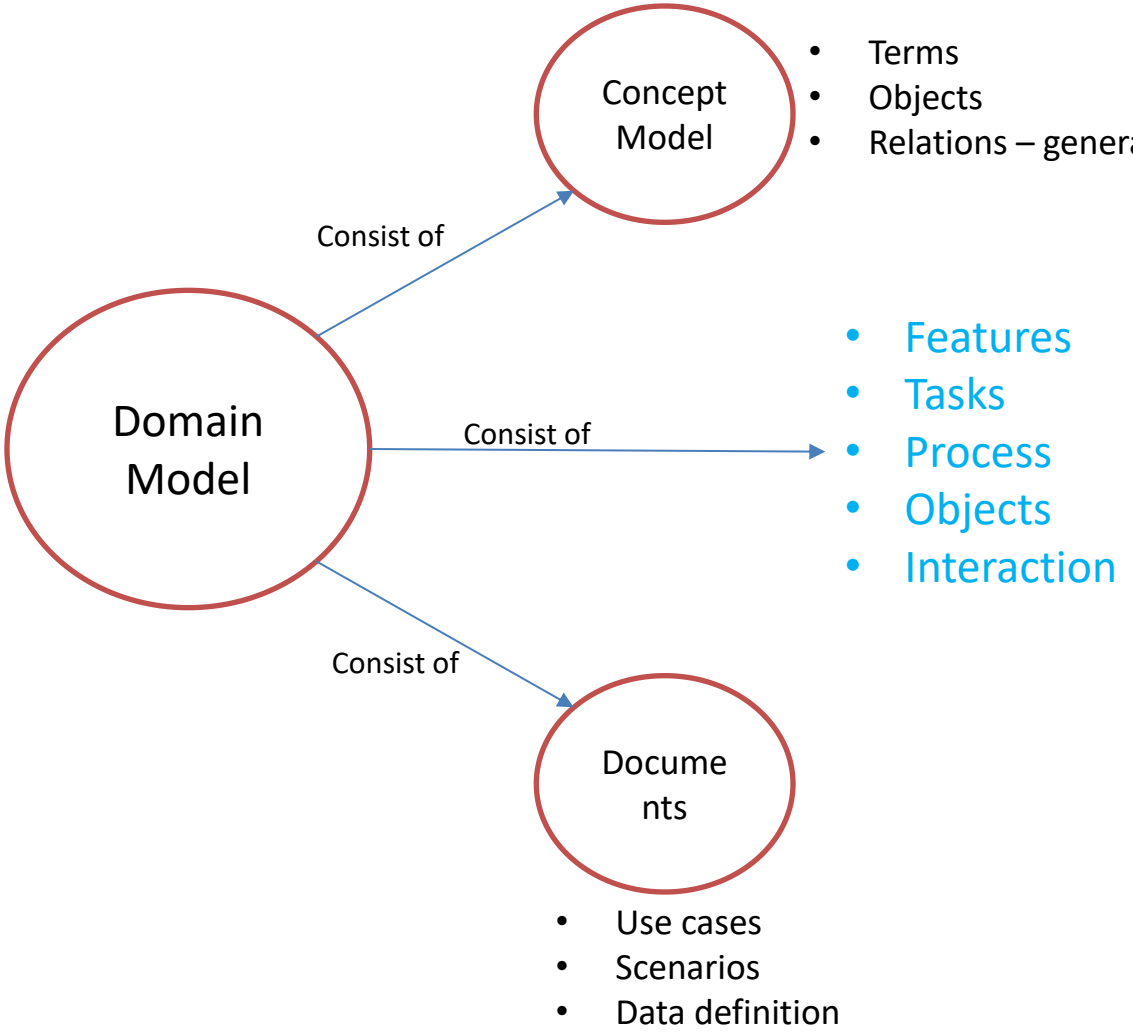
EXAMPLE



E-Payment /Financial

Add new subscriber (MBJB, Celcom, Maxis)

APPLICATION SOFTWARE AND DOMAIN



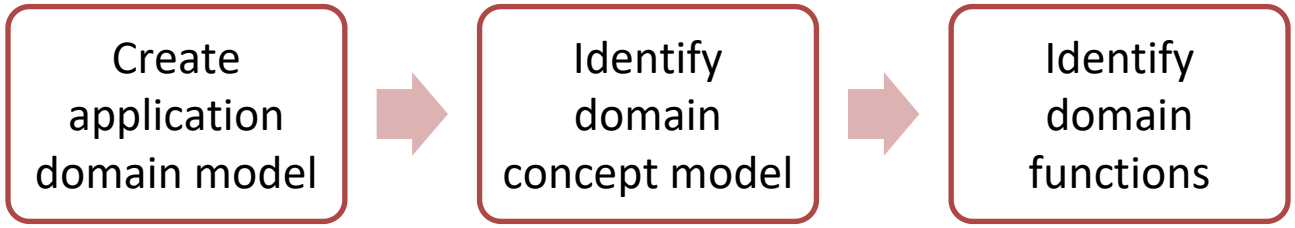
- Terms
- Objects
- Relations – generalization, composition, services

- Features
- Tasks
- Process
- Objects
- Interaction

- Use cases
- Scenarios
- Data definition

elements	examples
Features	Online payment
Tasks	Pay bills
Process	1. Click favorite 2. Select bill payments 3. Click Pay Bills 4. Select Astro
Object	Astro, Account
Interaction	Composition, generalization

Main Task Domain Modeling



IDENTIFY DOMAIN CONCEPTS



- Things, places, person
- Domain experts

- Payee, Payer, Account
- Boost owner
- Financial Manager

- Requirement documents
- Use cases
- Interviews
- Observation

- SRS, General Specification
- Formal interview with Boost stakeholders

- Read documents
- Conduct interview
- Conduct workshop
- Identify object
- Identify relationship
- Use noun-verb

NOUN ANALYSIS - STEPS

Existing resources

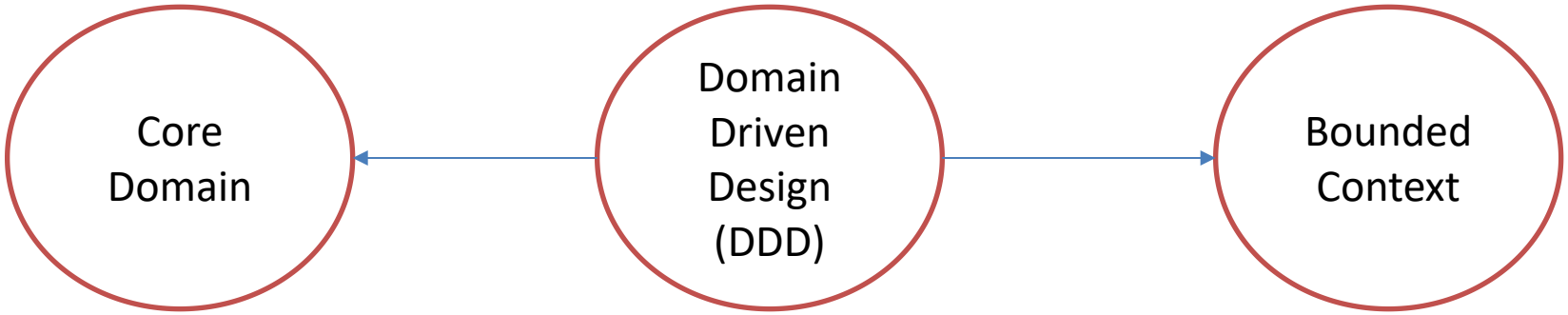
SRS/web page/domain experts/any documents

List all nouns	Fax, event, calendar event, calendar, appointment calendar, start time, access level, notification, notification mechanism, event collision
Remove irrelevant nouns	~~Fax~~, event, calendar event, calendar, appointment calendar, start time, access level, notification, notification mechanism, event collision
Remove instance	~~Fax~~, event, ~~calendar event~~, calendar, appointment calendar, start time, access level, notification, notification mechanism, event collision
Remove synonyms	~~Fax~~, event, ~~calendar event~~, calendar, ~~appointment calendar~~, start time, access level, notification, notification mechanism, event collision
Identify properties	~~Fax~~, event, ~~calendar event~~, calendar, ~~appointment calendar~~, ~~start time~~, ~~access level~~, notification, notification mechanism, event collision
Identify services	~~Fax~~, event, ~~calendar event~~, calendar, ~~appointment calendar~~, ~~start time~~, ~~access level~~, notification, ~~notification mechanism~~, event collision
Remove related nouns	~~Fax~~, event, ~~calendar event~~, calendar, ~~appointment calendar~~, ~~start time~~, ~~access level~~, notification, ~~notification mechanism~~, ~~event collision~~

NOUN ANALYSIS - GUIDELINES

- Always identify nouns in the documents/interview
- Eliminate any irrelevant nouns
- Avoid COMPUTER IMPLEMENTATION nouns
- Identify concepts name closely with application domain

DOMAIN DRIVEN DESIGN



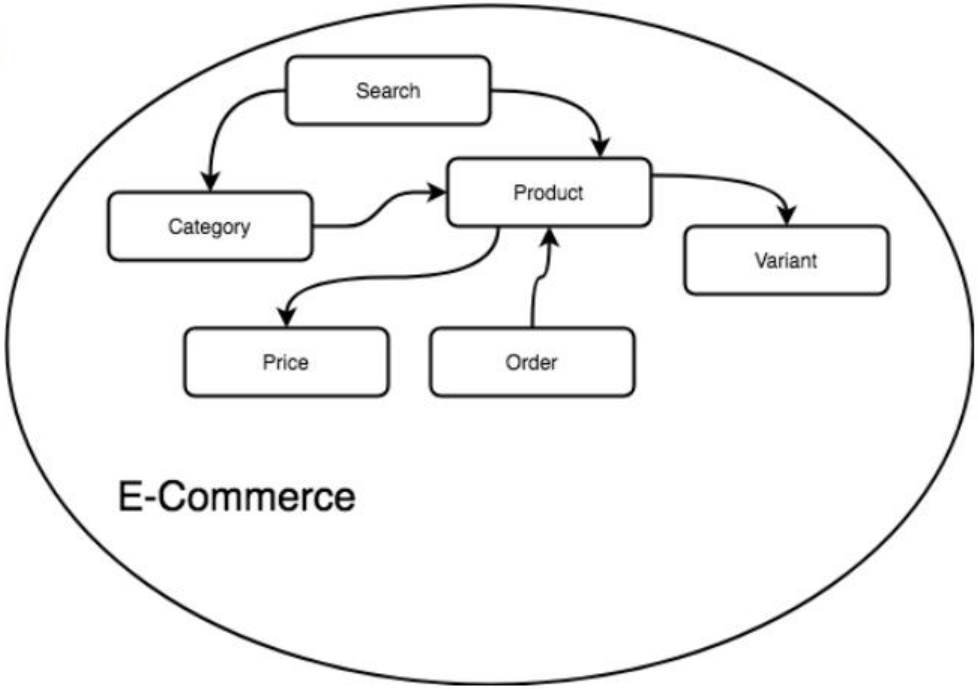
- Breaking down into multiple domain - Sub-domain
- Sub-Domain (Generic and Supporting)
- Business oriented focus

- Components and its relation
- 1 bounded context = 1 concept

- All domain must be connected through context mapping
- Context mapping – Relation between domain
- Context mapping
  - Restful API
  - Web services
  - JSON, XML
  - Microservices

Bounded Context: E-Commerce

- Example Concepts
  - Search in a Category
  - Select a Product
  - Choose Variant
  - Find Price
  - Place Order



DEVELOP AND REFINE AN UBIQUITOUS LANGUAGE

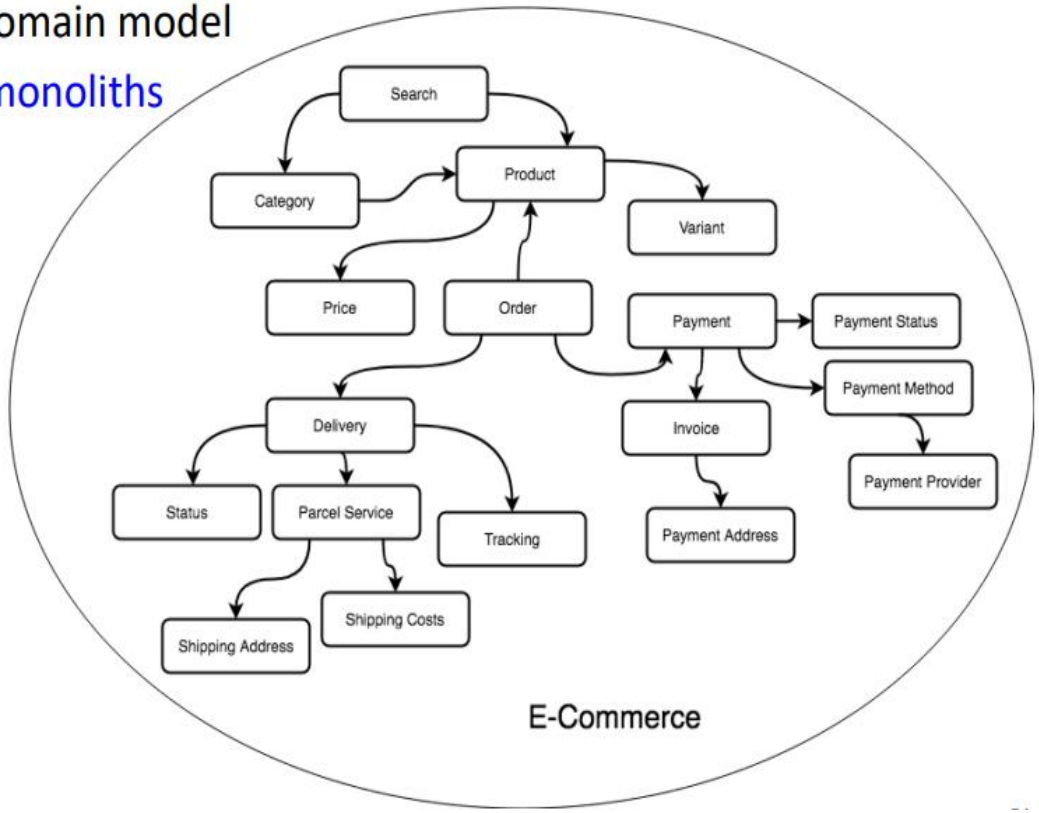
- Start with the nouns
  - representing the elements of your model
- Write scenarios describing the interaction of model elements
  - Textual
  - Informal

"A product is always assigned to one or more categories. A customer can navigate through a category to find and select a product he intends to buy. The product will then be added to the shopping cart of the customer"

- Extension: Behavior Driven Development

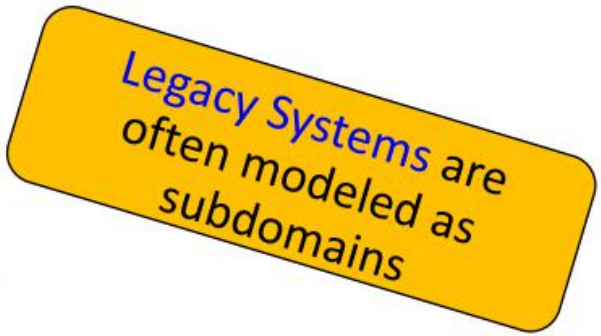
BOUNDED CONTEXT: BIG BALL OF MUD

- Anti-Pattern: Big Ball of Mud
  - Bounded context is stuffed with to many concepts
  - One big domain model
  - Leads to monoliths

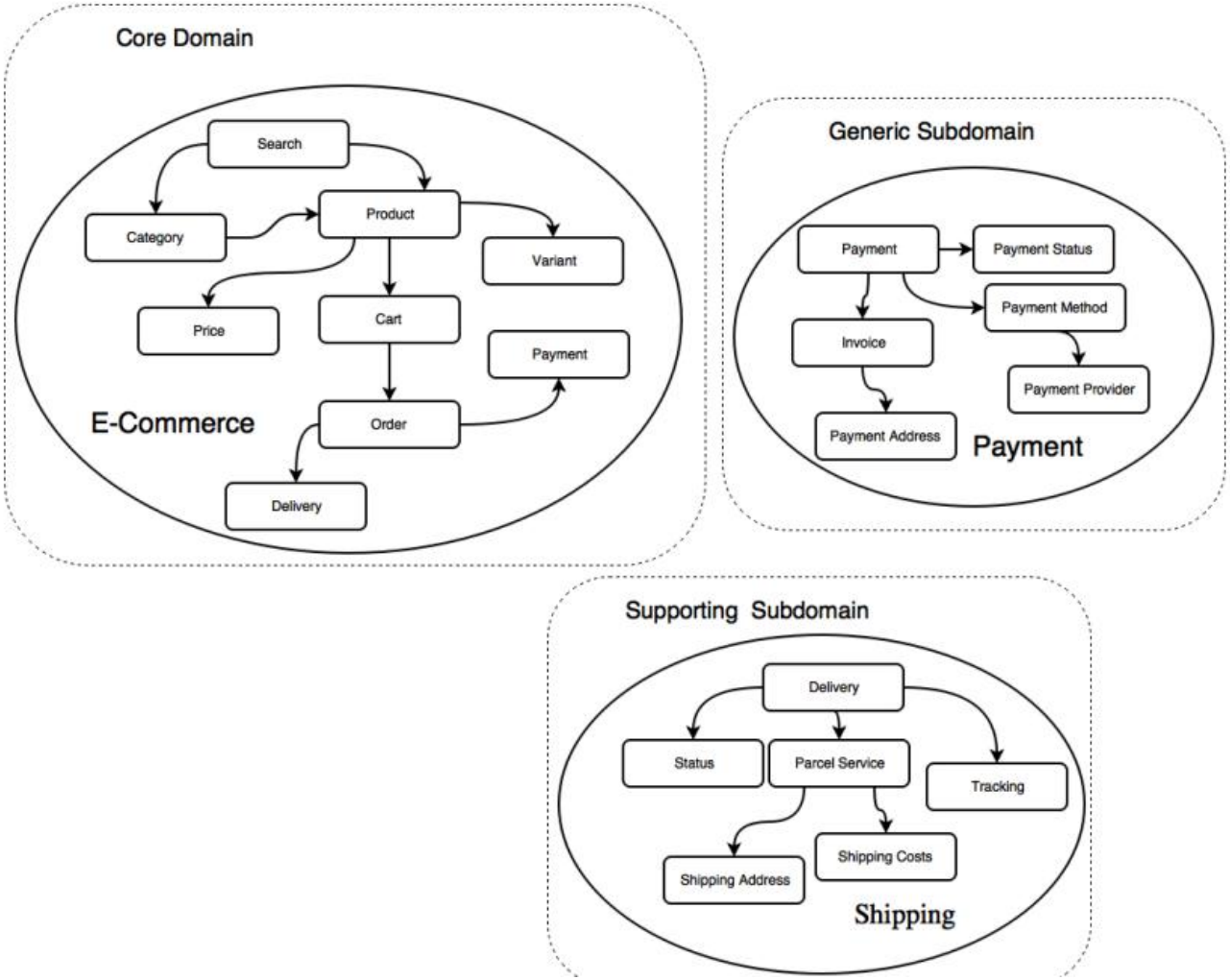


STRATEGIC DESIGN & SUBDOMAINS

- Subdomain: Part of your application or business domain
  - Each is modeled as a bounded context
  - Different types of subdomains
- Core Domain: central subdomain
- Supporting Subdomain
  - No real expertise
  - No standard software solution available
  - Needed to be modelled and implemented
  - Candidate for outsourcing
- Generic Subdomain
  - no special expertise needed: make or buy



STRATEGIC DESIGN & SUBDOMAINS

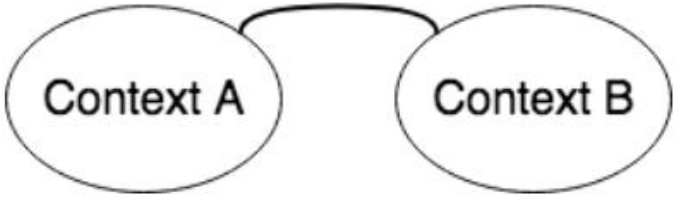


AMBIGUOUS CONCEPT

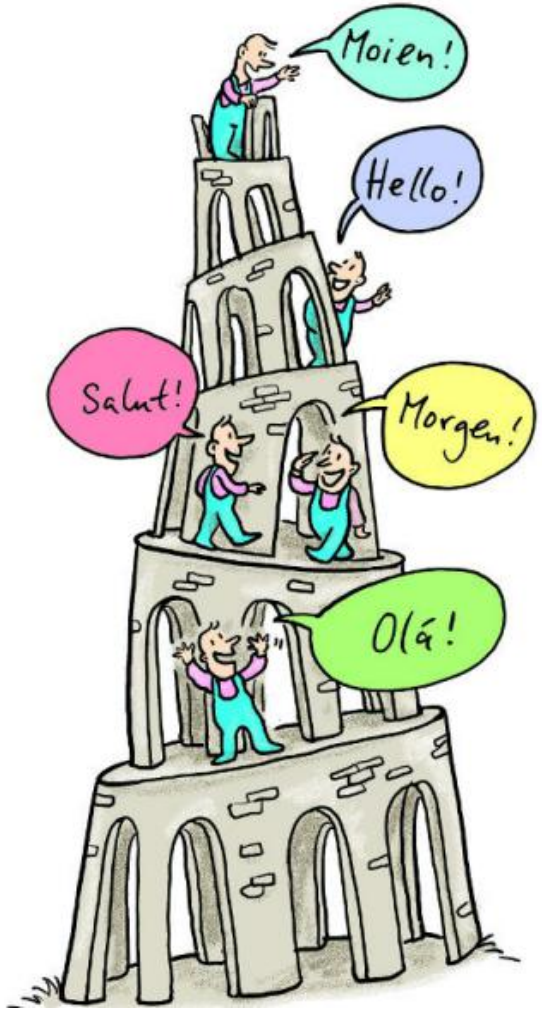
- Concepts can have different meanings in different contexts
- Example: Flight
  - Context: Ticketing
    - *A passenger travels from airport A to airport B with a layover at airport C with airline Y-Tours in seat 4H in the economy class.*
  - Context: Air Traffic Control
    - *An aircraft Y-1337 from airline Y-Tours starts from airport A and lands at airport C with flight number Y4711.*
- Naive solution for one ubiquitous language
  - TicketFlight
  - AirTrafficControlFlight

STRATEGIC DESIGN & CONTEXT MAP

- Context Map
  - Relation between two bounded contexts
- Each context has its own language

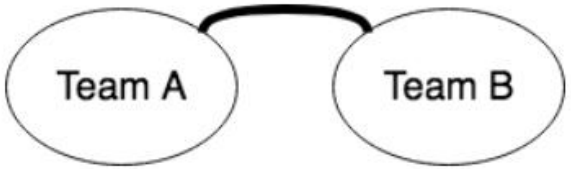


- Mapping solutions
  - Translate languages
  - Share languages

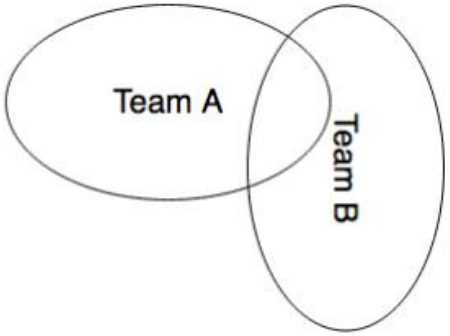


CONTEXT MAPPING

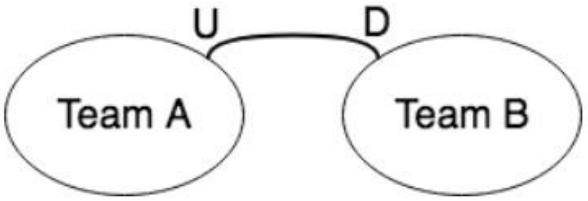
- Partnership
  - Both contexts using the same language
  - Hard to maintain
  - Apply a time limit



- Shared Kernel
  - Small intersection model
  - Shared code base

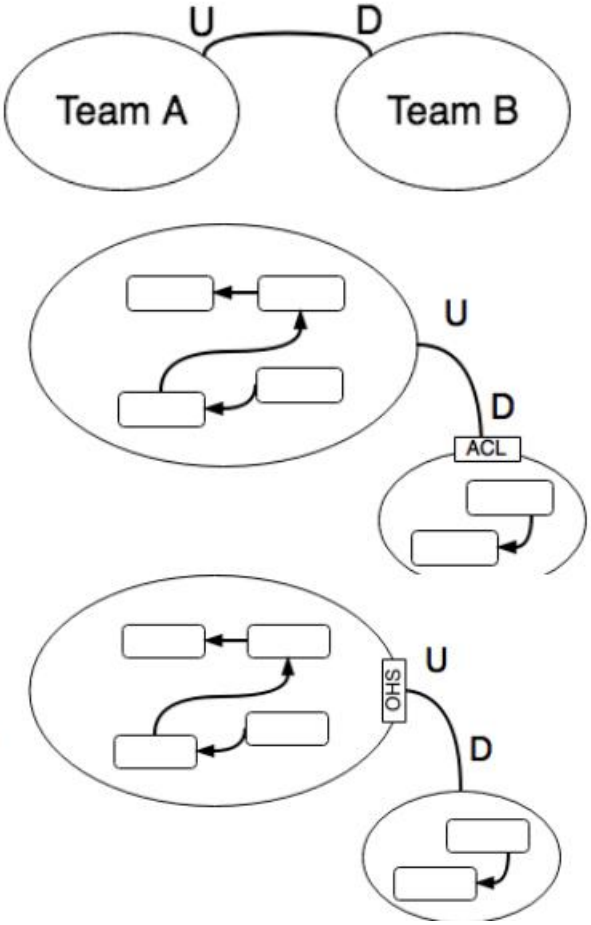


- Customer-Supplier
  - Supplier: Upstream (U)
  - Customer: Downstream (D)
  - Customer depends on Supplier



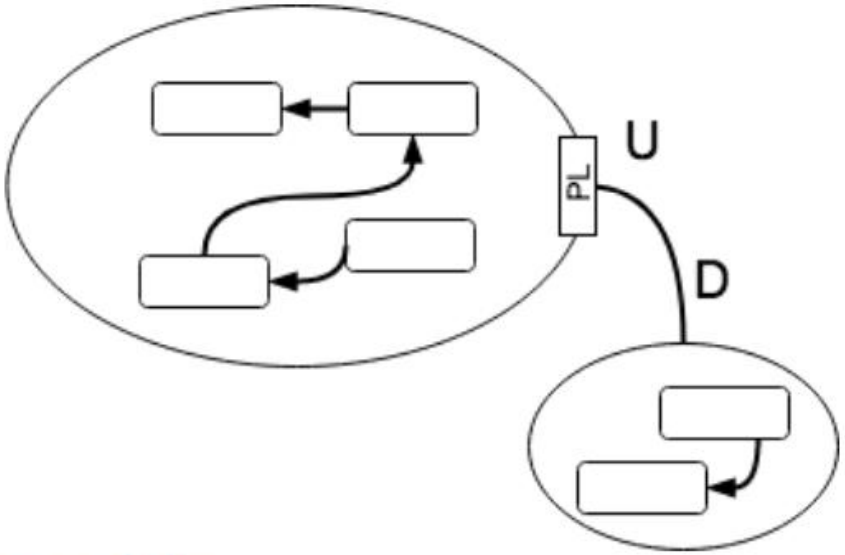
CONTEXT MAPPING

- Conformist
  - Customer adopts supplier language completely
  - Example: integrate external API
- Anti-Corruption Layer
  - Controlled integration
  - Protect from upstream model
  - Best practice
  - Patterns: Adapter
- Open Host Service
  - Context defines a simplified interface
  - Pattern: Facade



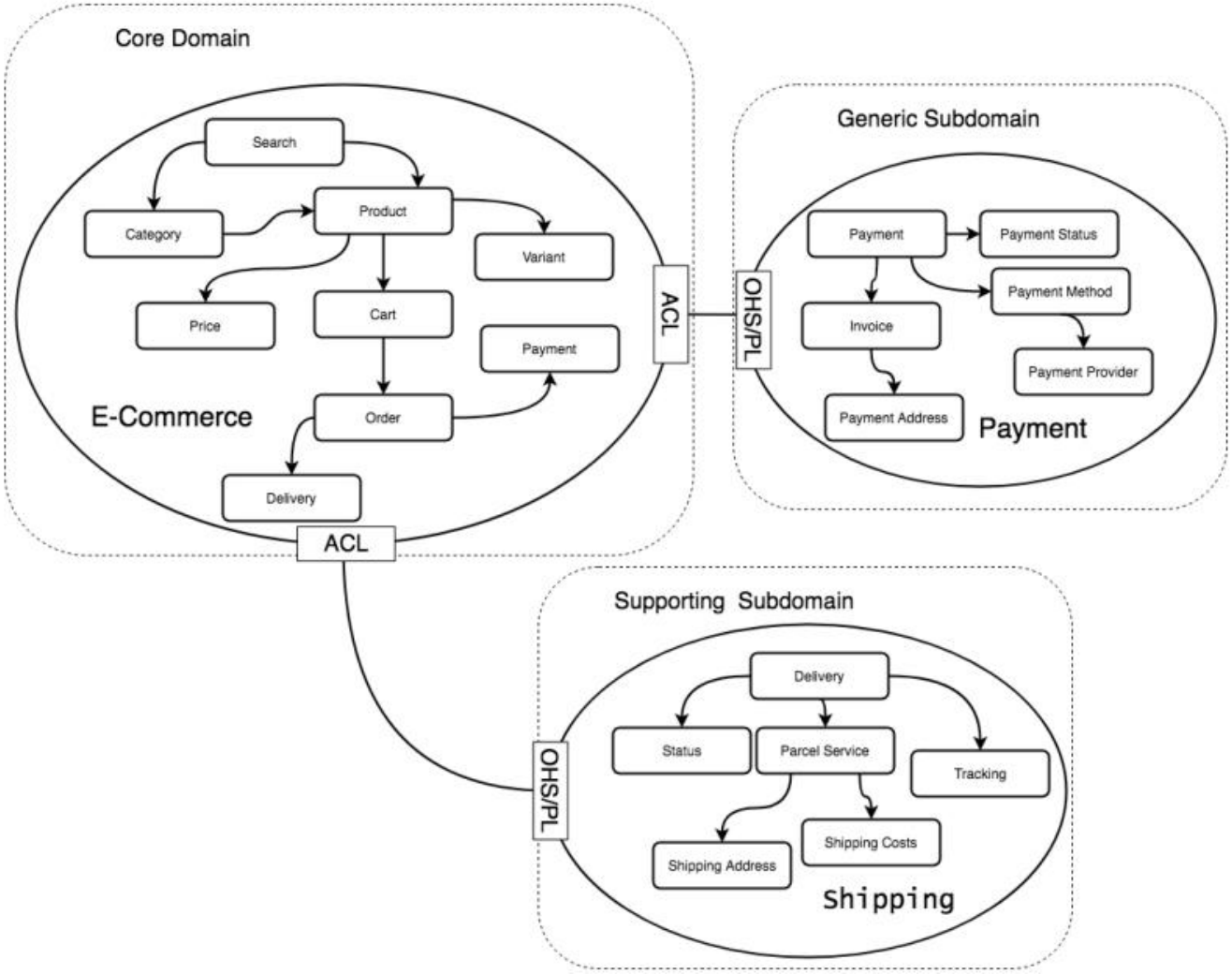
CONTEXT MAPPING

- Published Language
  - well-documented
  - simple & easy to use
  - JSON/XML-Schema
  - Techniques: REST,SOAP



- Central Question
  - Integrate or translate other models
- Context Mapping Patterns can be combined
  - Open Host Service provides a Published Language
  - Customer implements an Anticorruption Layer for a Published Language

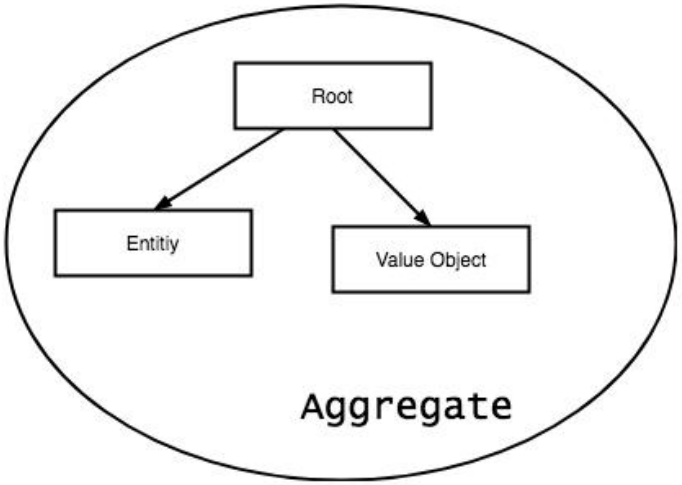
STRATEGIC DESIGN AND CONTEXT MAPPING



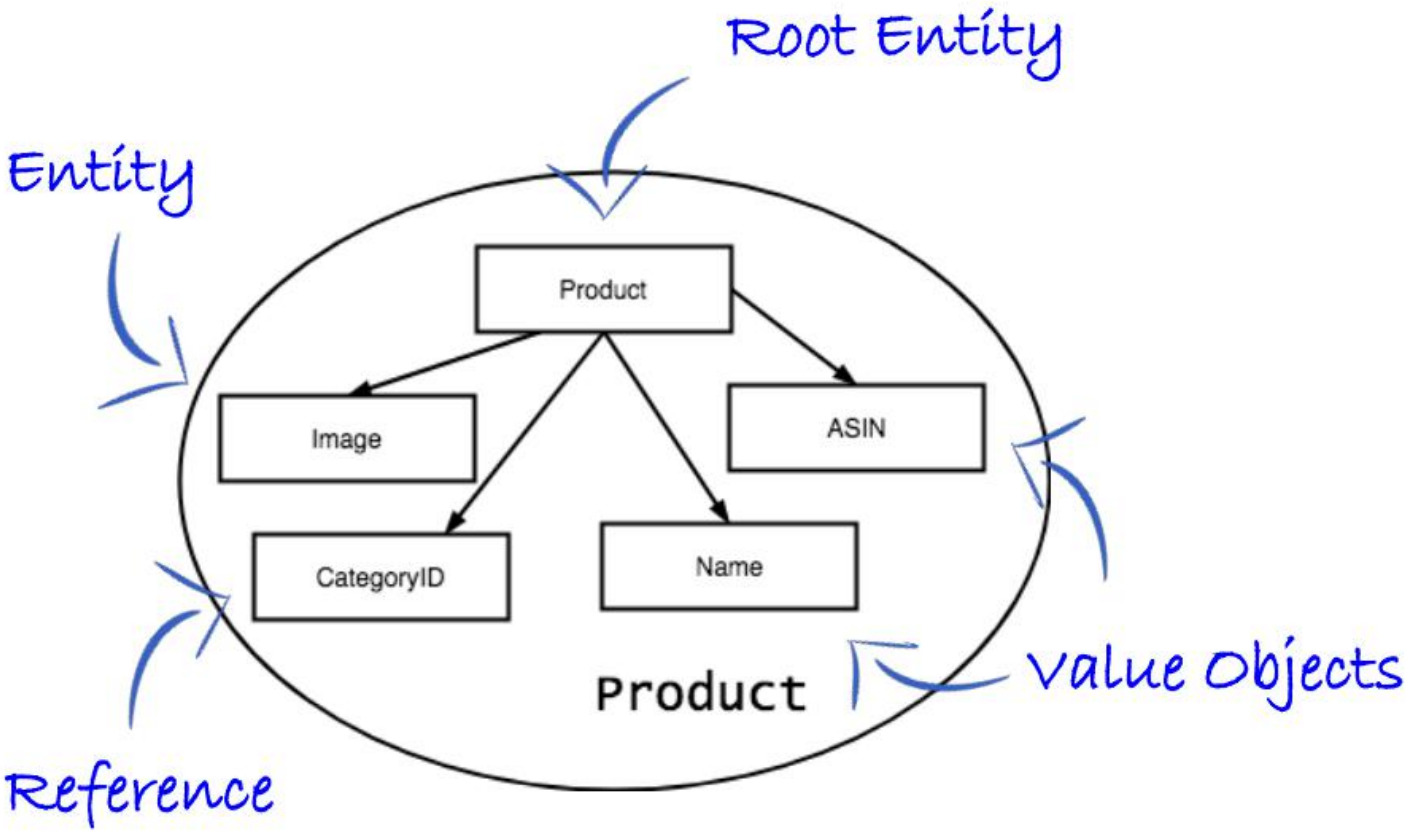
TACTICAL DESIGN & BOUNDED CONTEXT

- Concepts inside a bounded context can be complex
- Can be modelled as aggregates
- Aggregate
  - consists of:
    - Entities
    - Value Objects
    - Root Entity (name of the concept)
  - defines:
    - conceptual boundary
    - transactional boundary

Product
- Name
- Description
- Media (Images, Video, Audio)
- Category (Relation)
- Variant (Relation)



AGGREGATES

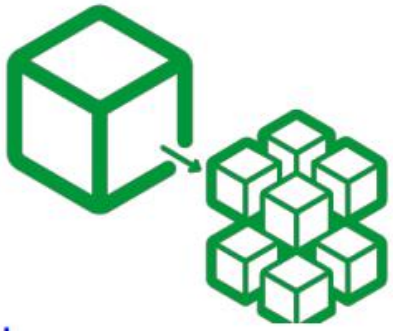

Root Entity
Entity
Product
Image
ASIN
CategoryID
Name
Product
Value Objects
Reference

DOMAIN DRIVEN DESIGN: ORGANIZATION

- Map teams to bounded contexts
  - One team per bounded context
  - One team can handle more than one context
  - Core domain: assign the best team
- Software Configuration Management
  - One repository for one bounded context
  - One database schema
  - A consistent set of acceptance tests/rules
- Ubiquitous Language
  - Terms defined inside a bounded context
  - Each team member knows, use and understands the ubiquitous language

Domain driven design: software architecture-preview

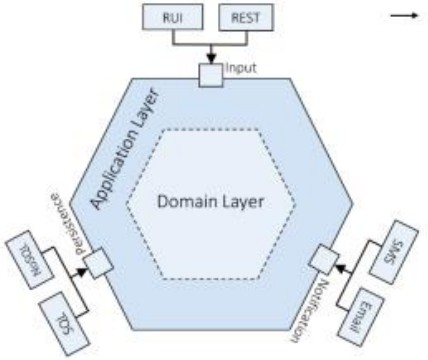
- Bounded Context
  - Architectural Style: Microservices



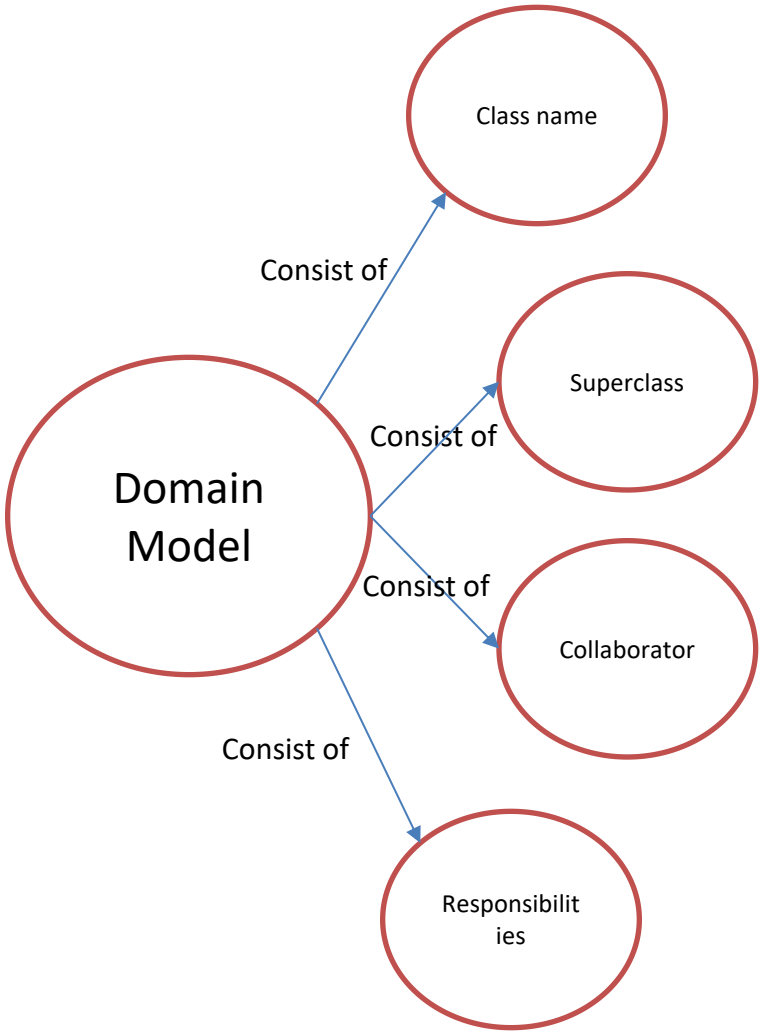
- Aggregates
  - OO-Anti-Pattern: Anemic Domain Model

- Context Mapping
  - RESTful-Webservices and SOA
  - Event-Driven Architecture
  - Message Driven Architecture (Messaging)
  - Hexagonal/Ports and Adapter

{ REST }



CLASS RESPONSIBILITY COLLABORATOR



Use case

The lecturer enters the name of the new course and classifies it according to one of the predefined course types. Then he chooses a lecture hall and assigns it to the course. If he wants an examination for the course he registers the examination and assigns respective examiners.

Lecturer	
Super Classes:	
Description: A lecturer is allowed to teach students and register courses	
Attributes:	
Name	Description
name	
organization	
Responsibilities:	
Name	Collaborator
create a Course	Course
register Examination	Examination
book a Lecture Hall	Lecture Hall

Lecture Hall	
Super Classes:	
Description:	
Attributes:	
Name	Description
Responsibilities:	
Name	Collaborator
check availability	
prepare weekly schedule	

CRC cards - Structure

- Class
  - Name (identifier) of the class (term)
- Superclass
  - Generalized terms (if already known)
- Responsibilities
  - Services that are offered
- Collaborators
  - Terms that are related concerning the usage of offered services
  - Used to realize own services





SUMMARY



- Domain Driven Design:
  - Bounded Context: Identify semantic isolated areas inside application domain
  - Ubiquitous Language: Clear consistent definition of concepts
  - Context Mapping: Define relation between different contexts
  - Aggregates: Define consistent Entities
  - Stakeholders: Domain Experts & Developers
- Advantages
  - Easy to learn, hard to master
  - Technology independent
  - Informal representation (no UML required)

SUMMARY

- A major goal of analysis is to create a domain model.
  - Use cases
  - Domain concept model



- Concepts and terms are identified based on existing information (documentation)
  - UML class diagrams, glossary, CRC cards
- Activities have to be performed iteratively.

